

Individual Reflection on Group Project

Project Title: Edge-Deployable Rock Detection and Burial Status Classification

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Team / Group: Team 9

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1. Your views on the completing the project as a team

Teamwork was appropriate for this project, as it involved preparing the data sets, training the detectors, converting formats and hardware measurement. I was tasked with deploying Raspberry Pi 5 and analysing the performance of the hardware for three models of the selected YOLOv8n detector: PyTorch FP32, TFLite FP32 and NCNN FP32. I also participated in setting up the cross-format comparison, reporting accuracy, efficiency and resources needed consistently. Since my work was towards the end of the dependency chain, the measurements were only as accurate as the detector and conversions that were handed to me, and the team's decision was based on the baseline I made. Benchmarking was found to be a shared responsibility, and the reproducible conditions were as important as the numbers.

2. Your views on project management and execution of your team

Since there was only one shared device, access was scheduled and each measurement followed one protocol: batch size 1, 640×640 letterboxed inputs, confidence 0.25, IoU 0.70, 3 warmup, 1 timed pass over the same 90 images containing 135 labelled rocks. We recorded the intended protocol of 10 warm ups, three timed passes and a 30-minute stability test separately from the one performed, rather than presenting the plan as evidence. An early NCNN FP32 attempt rebooted the Pi on a non-official power adapter; the benchmark completed after switching to the official 27 W supply.

3. Reflections on Successes, Failures, and Retrospective Changes

- **What Worked Well:** Benchmarking three full-precision formats separated backend implementation from numerical precision. The mean latency dropped from 397.48 ms for PyTorch FP32 to 143.70 ms for TFLite FP32 and 75.09 ms for NCNN FP32 at the same precision, so much of the edge speed-up is due to the inference engine before quantisation. NCNN FP32 reached 13.26 FPS but at 83.32% mean CPU against 44.47% for PyTorch.
- **What Did Not Work Well:** NCNN FP32 provided the best trade-off between accuracy and speed of my three formats, but hard-reset the Pi at a later point in a continuous run, without leaving any evidence of sustained operation. One timed pass gave no run-to-run variance, and without an Active Cooler my maximums of 61.70 °C and 66.10 °C are not equilibrium temperatures.
- **What I Would Do Differently:** I would run repeated timed passes to report variance, and log CPU frequency and throttle flags next to the temperature and apply the continuous stability test to all formats, not just the winner.

4. Any other reflections

This was a very important lesson — the difference between a format that runs and a format that is deployable: export, single-image inference, a complete 90-image benchmark and sustained operation are four different levels of evidence. No attempt was made to estimate incomplete outcomes; these were reported as incomplete. There are also limits to runtime optimisation, since the half-buried weakness existed before conversion and has not been fixed by the speed-up.